Table of Contents

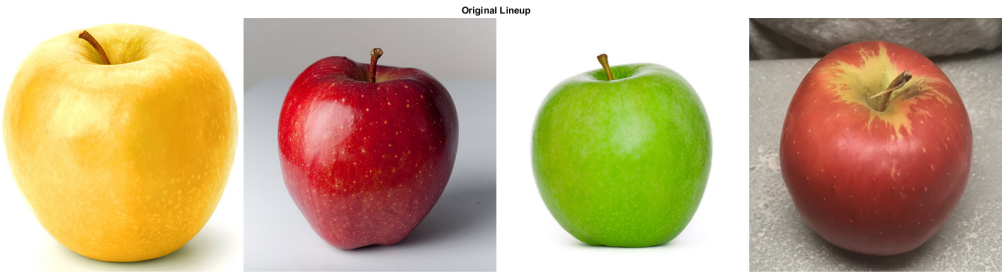
Final Project 1
Part 1 1
Part 2 2
Stephen Donchez 2
Matthew Ballas 3
James DeLaurentis 4
Nick Florio 5
Part 3 6
Stephen Donchez 6
Matt Ballas 7
James DeLaurentis 8
Nick Florio 9
Part 4 10
Stephen Donchez 10
Matt Ballas 11
James DeLaurentis 13
Nick Florio 14
Part 5 16

Final Project

```
%ECE 2409
%Group 6
%Matt Ballas, James Delaurentis, Stephen Donchez, Nick Florio
```

Part 1

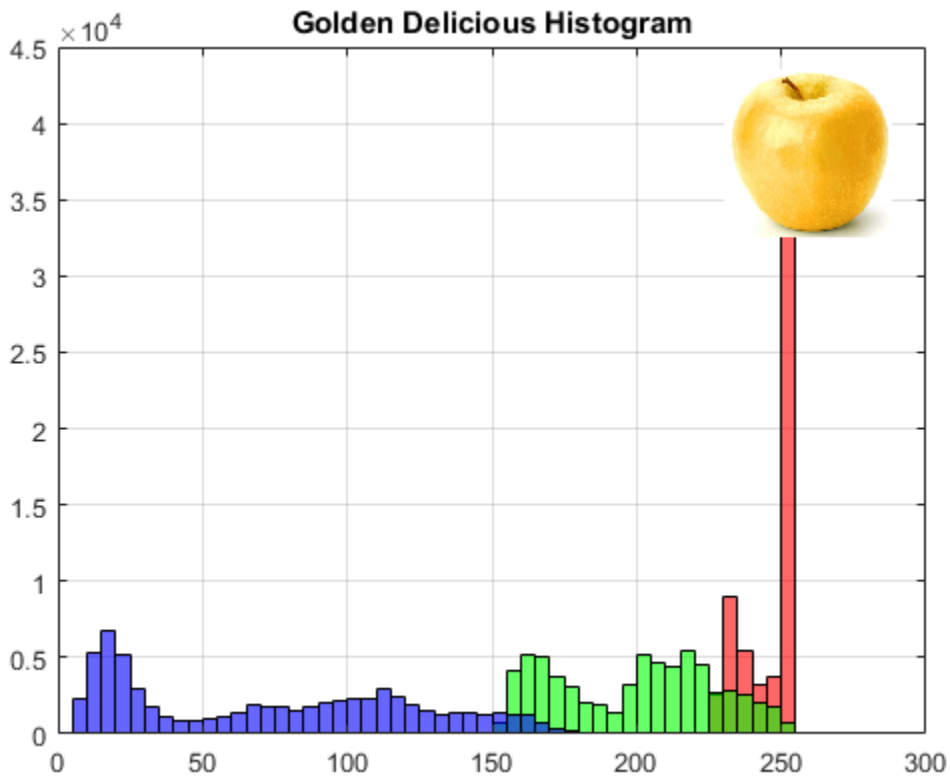
```
GDi=imread('golden-delicious-apple.jpg');
HCi=imread('honeycrisp.png');
HCir=imresize(HCi,[400 400]);
GSi=imread('GrannySmith.jpg');
GSir=imresize(GSi,[400 400]);
RDi=imread('RedDelicious.jpg');
RDir=imresize(RDi,[400 400]);
T=[GDi RDir GSir HCir];
imshow(T);
title('Original Lineup');
```



Part 2

Stephen Donchez

```
figure
GD=GDi(70:315,68:340,:);
rGD=GD(:,:,1);
gGD=GD(:,:,2);
bGD=GD(:,:,3);
histogram(rGD,'facecolor','r','binwidth',5);
grid; hold on;
histogram(gGD,'facecolor','g','binwidth',5);
histogram(bGD,'facecolor','b','binwidth',5);
title('Golden Delicious Histogram');
axes('pos',[.7 .7 .2 .2])
imshow(GDi)
%
%
%
%
%
%
```

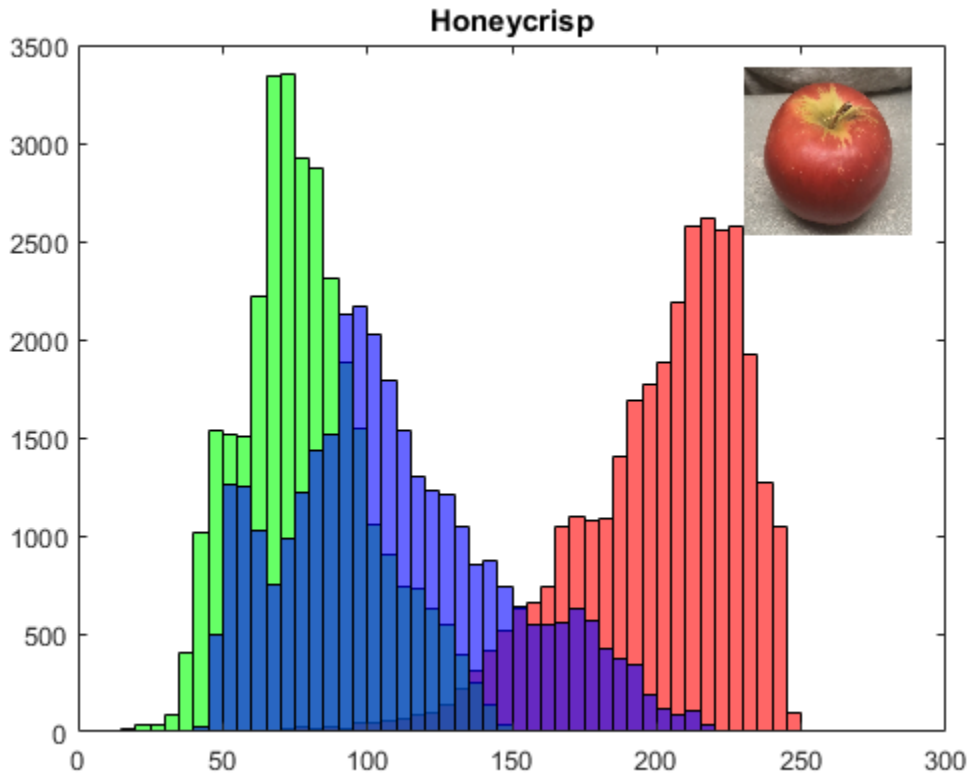


Matthew Ballas

```
honey = HCir(84:258,125:307,:);
r=honey(:,:,1);
b=honey(:,:,2);
g=honey(:,:,3);

figure
histogram(r,'facecolor','r','binwidth',5);
hold
histogram(g,'facecolor','g','binwidth',5);
histogram(b,'facecolor','b','binwidth',5);
title('Honeycrisp');
axes('pos',[.7 .7 .2 .2])
imshow(HCir);
%
%
%
%
%
%
```

Current plot held

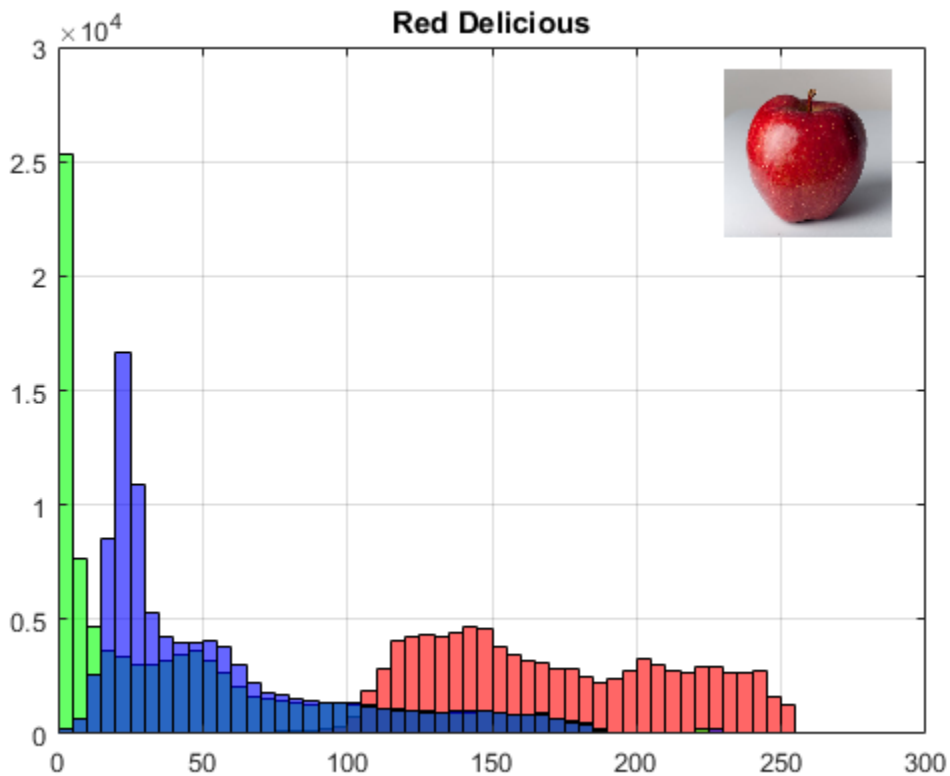


James DeLaurentis

```
rd=RDi(148:471,126:418,:);
rrd1=rd(:,:,1);
grd1=rd(:,:,2);
brd1=rd(:,:,3);

figure
histogram(rrd1,'facecolor','r','binwidth',5)
hold
histogram(grd1,'facecolor','g','binwidth',5)
histogram(brd1,'facecolor','b','binwidth',5)
grid
title('Red Delicious')
axes('pos',[.7 .7 .2 .2])
imshow(RDi)
%
%
%
%
%
%
```

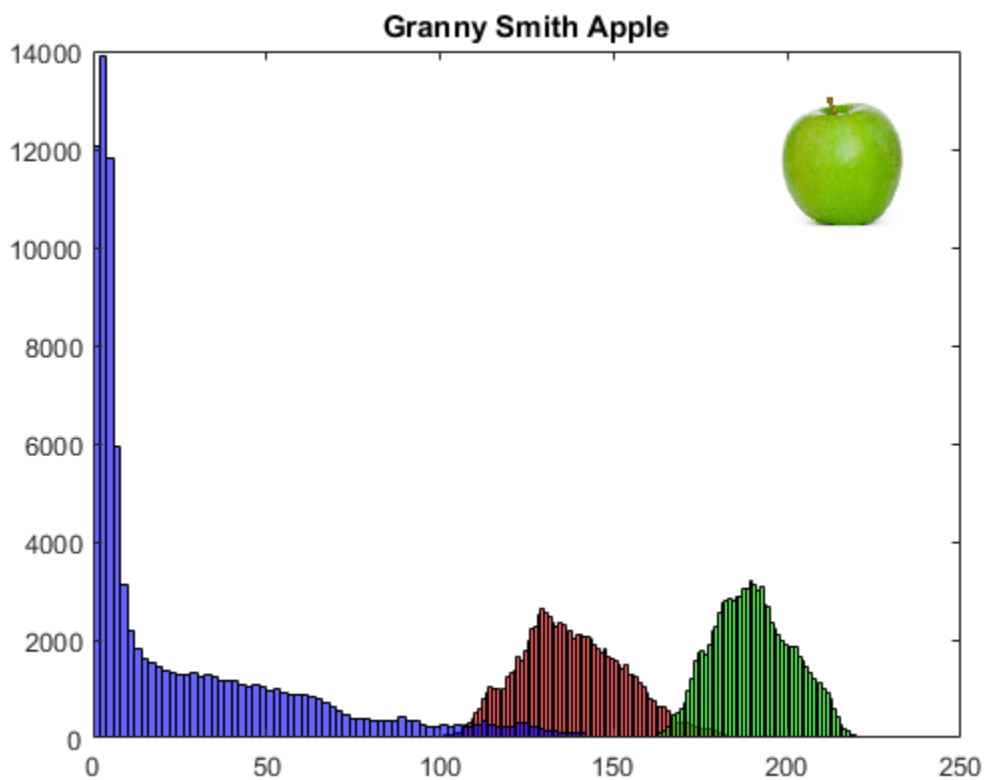
Current plot held



Nick Florio

```
Greeny= GSi(178:467,149:465,:);
R=Greeny(:,:,1);
G=Greeny(:,:,2);
B=Greeny(:,:,3);
figure
histogram(R, 'facecolor', 'r')
hold
histogram(G, 'facecolor', 'g')
histogram(B, 'facecolor', 'b')
title('Granny Smith Apple')
axes('pos',[.7 .7 .2 .2])
imshow(GSir);
%
%
%
%
%
%
```

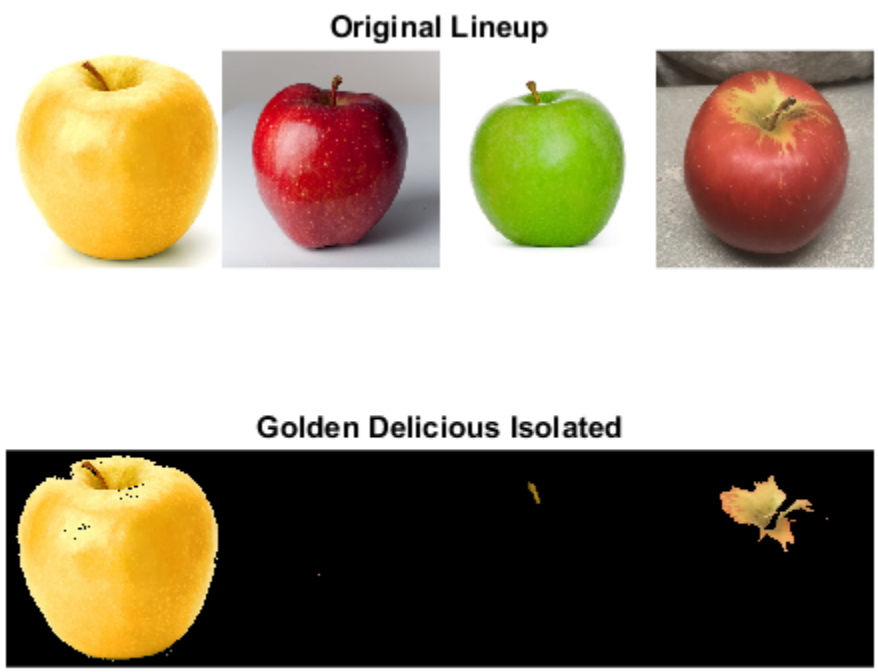
Current plot held



Part 3

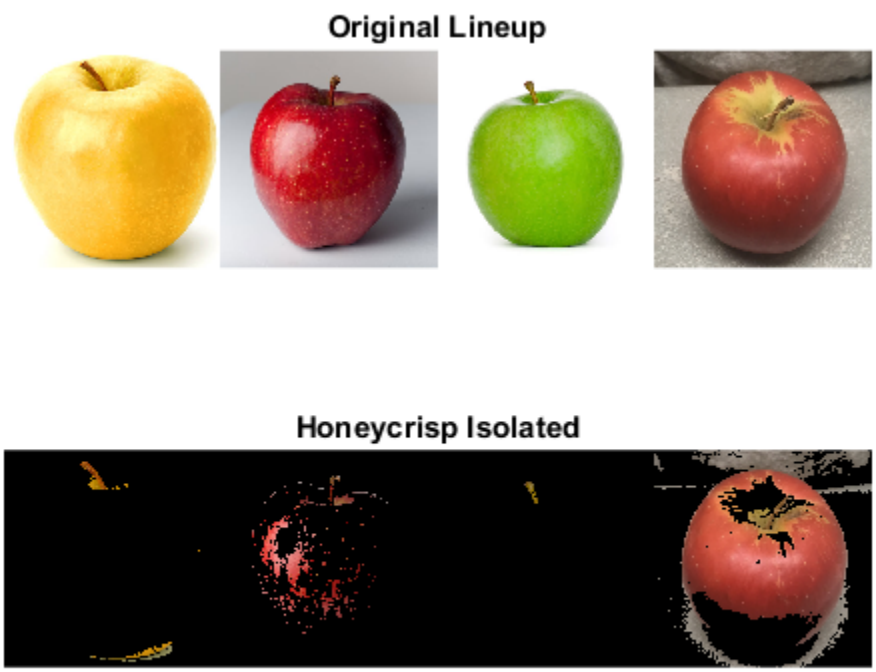
Stephen Donchez

```
rT=T(:,:,1);
gT=T(:,:,2);
bT=T(:,:,3);
GDSelect=(bT+50<gT & gT<rT & bT<175);
GDSelectInt=uint8(GDSelect);
GDShow=cat(3,GDSelectInt.*T(:,:,1),GDSelectInt.*T(:,:,2),GDSelectInt.*T(:,:,3));
figure;
subplot(2,1,1)
imshow(T);
title('Original Lineup');
subplot(2,1,2)
imshow(GDShow);
title('Golden Delicious Isolated');
%
%
%
%
%
%
```



Matt Ballas

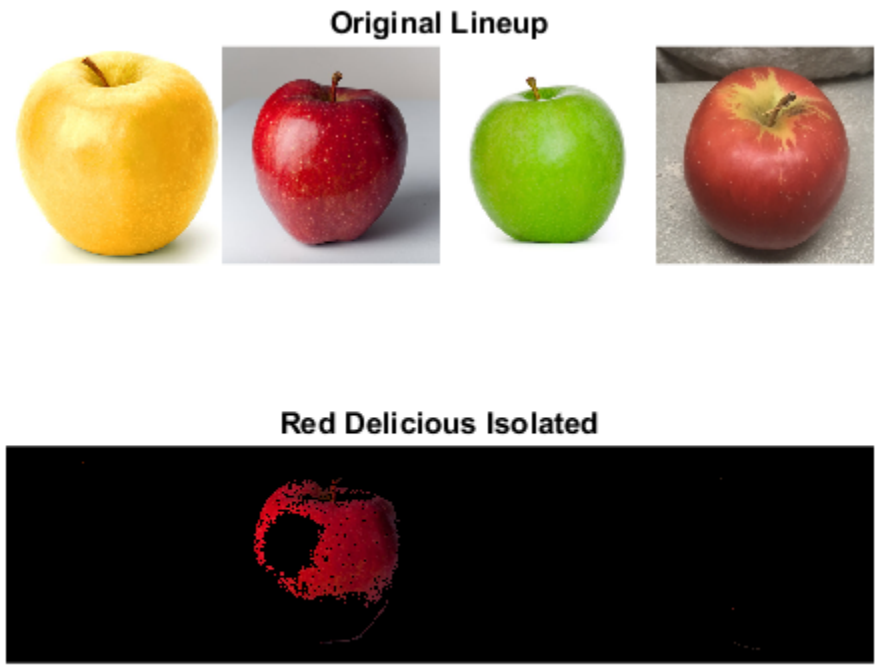
```
r=T(:,:,1);
g=T(:,:,2);
b=T(:,:,3);
i = ( r>105 & b-g==0 & b<200 & g<150 & r>g);
r(~i)=0;
b(~i)=0;
g(~i)=0;
foundhoney=cat(3,r,g,b);
figure
subplot(2,1,1)
imshow(T);
title('Original Lineup');
subplot(2,1,2)
imshow(foundhoney);
title('Honeycrisp Isolated');
%
%
%
%
%
%
```



James DeLaurentis

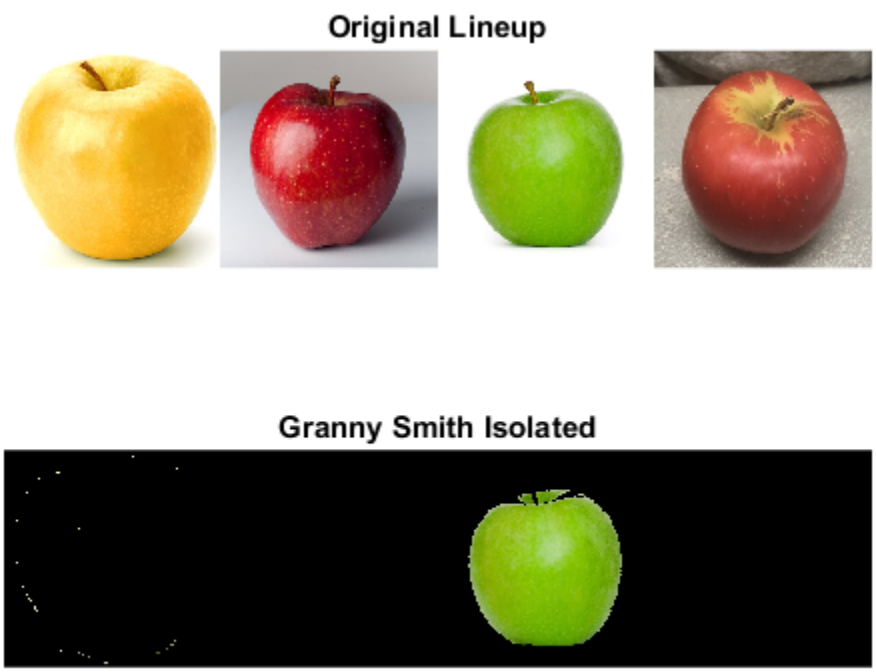
```
figure
rrd=(T(:,:,1));
grd=(T(:,:,2));
brd=(T(:,:,3));

prd=uint8((rrd>grd & rrd>brd & grd<35));
PRD=prd.*rrd;
PGD=prd.*grd;
PBD=prd.*brd;
ORD=(PRD);
ORD(:,:,2)=(PGD);
ORD(:,:,3)=(PBD);
%ORD1=uint8(ORD);
subplot(2,1,1)
imshow(T);
title('Original Lineup');
subplot(2,1,2)
imshow(ORD);
title('Red Delicious Isolated');
%
%
%
%
```



Nick Florio

```
R2=T(:,:,1);
G2=T(:,:,2);
B2=T(:,:,3);
Greeny=(G2>R2 & G2>B2 & R2>B2+25);
GSSelectInt=uint8(Greeny);
GSShow=cat(3,GSSelectInt.*T(:,:,1),GSSelectInt.*T(:,:,2),GSSelectInt.*T(:,:,3));
figure
subplot (2,1,1);
imshow(T);
title('Original Lineup');
subplot (2,1,2);
imshow(GSShow);
title('Granny Smith Isolated');
%
%
%
%
%
%
%
%
```



Part 4

```
T=[GSir GDi HCir RDir];
```

Stephen Donchez

```
rT=T(:,:,1);
gT=T(:,:,2);
bT=T(:,:,3);
GDSelect=(bT+50<gT & gT<rT & bT<175);
GDSelectInt=uint8(GDSelect);
GDShow=cat(3,GDSelectInt.*T(:,:,1),GDSelectInt.*T(:,:,2),GDSelectInt.*T(:,:,3));
figure;
subplot(2,1,1)
imshow(T);
title('Original Lineup');
subplot(2,1,2)
imshow(GDShow);
title('Golden Delicious Isolated');

[Tx, Ty, Tz]=size(T);
GDSelect1=GDSelect(:,1:(Ty/4));
GDSelect2=GDSelect(:,(Ty/4):(Ty/2));
GDSelect3=GDSelect(:,(Ty/2):(3*Ty/4));
GDSelect4=GDSelect(:,(3*Ty/4):Ty);
GDSum1=sum(GDSelect1(:));
GDSum2=sum(GDSelect2(:));
GDSum3=sum(GDSelect3(:));
GDSum4=sum(GDSelect4(:));
if GDSum1>GDSum2 && GDSum1>GDSum3 && GDSum1>GDSum4
    fprintf('This apple is the leftmost apple\n')
else
    if GDSum2>GDSum3 && GDSum2>GDSum4
        fprintf('This apple is the second left apple\n')
    else
        if GDSum3>GDSum4
            fprintf('This apple is the third from left apple\n')
        else
            fprintf('This apple is the fourth apple from the left\n')
        end
    end
end
```

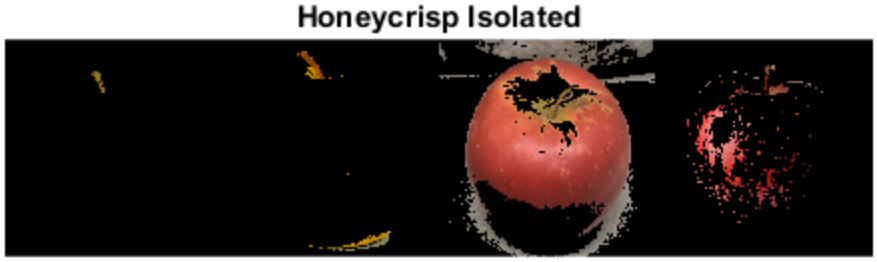
```
This apple is the second left apple
```

Matt Ballas

```
r=T(:,:,1);
g=T(:,:,2);
b=T(:,:,3);
i = ( r>105 & b-g==0 & b<200 & g<150 & r>g);
r(~i)=0;
b(~i)=0;
g(~i)=0;
foundhoney=cat(3,r,g,b);
figure
subplot(2,1,1)
imshow(T);
title('Original Lineup');
subplot(2,1,2)
imshow(foundhoney);
title('Honeycrisp Isolated');
HCSelect1=i(:,1:(Ty/4));
HCSelect2=i(:,(Ty/4):(Ty/2));
HCSelect3=i(:,(Ty/2):(3*Ty/4));
HCSelect4=i(:,(3*Ty/4):Ty);
HCSum1=sum(HCSelect1(:));
HCSum2=sum(HCSelect2(:));
HCSum3=sum(HCSelect3(:));
HCSum4=sum(HCSelect4(:));
```

```
if HCSum1>HCSum2 && HCSum1>HCSum3 && HCSum1>HCSum4
    fprintf('This apple is the leftmost apple\n')
else
    if HCSum2>HCSum3 && HCSum2>HCSum4
        fprintf('This apple is the second left apple\n')
    else
        if HCSum3>HCSum4
            fprintf('This apple is the third from left apple\n')
        else
            fprintf('This apple is the fourth apple from the left\n')
        end
    end
end
%
%
%
%
%
%
%
%
%
```

This apple is the third from left apple



James DeLaurentis

```
figure
rrd=(T(:,:,1));
grd=(T(:,:,2));
brd=(T(:,:,3));

prd=uint8((rrd>grd & rrd>brd & grd<35));

PRD=prd.*rrd;
PGD=prd.*grd;
PBD=prd.*brd;
ORD=(PRD);
ORD(:,:,2)=(PGD);
ORD(:,:,3)=(PBD);
%ORD1=uint8(ORD);
subplot(2,1,1)
imshow(T);
title('Original Lineup');
subplot(2,1,2)
imshow(ORD);
title('Red Delicious Isolated');

RDSelect1=prd(:,1:(Ty/4));
RDSelect2=prd(:,(Ty/4):(Ty/2));
RDSelect3=prd(:,(Ty/2):(3*Ty/4));
RDSelect4=prd(:,(3*Ty/4):Ty);
RDSum1=sum(RDSelect1(:));
RDSum2=sum(RDSelect2(:));
RDSum3=sum(RDSelect3(:));
RDSum4=sum(RDSelect4(:));
if RDSum1>RDSum2 && RDSum1>RDSum3 && RDSum1>RDSum4
    fprintf('This apple is the leftmost apple\n')
else
    if RDSum2>RDSum3 && RDSum2>RDSum4
        fprintf('This apple is the second left apple\n')
    else
        if RDSum3>RDSum4
            fprintf('This apple is the third from left apple\n')
        else
            fprintf('This apple is the fourth apple from the left\n')
        end
    end
end
```

This apple is the fourth apple from the left

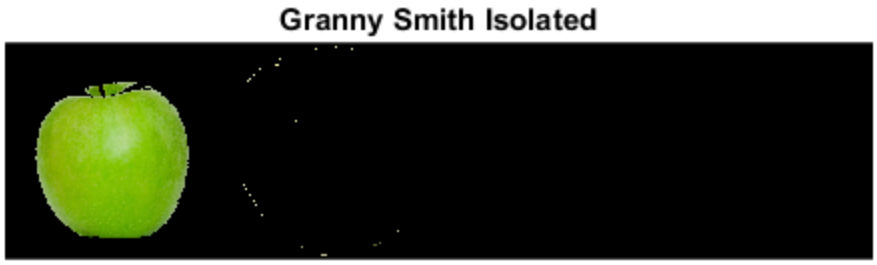
Nick Florio

```
R2=T(:,:,1);
G2=T(:,:,2);
B2=T(:,:,3);
Greeny=(G2>R2 & G2>B2 & R2>B2+25);
GSSelectInt=uint8(Greeny);
GSShow=cat(3,GSSelectInt.*T(:,:,1),GSSelectInt.*T(:,:,2),GSSelectInt.*T(:,:,3));
figure
subplot (2,1,1);
imshow(T);
title('Original Lineup');
subplot (2,1,2);
imshow(GSShow);
title('Granny Smith Isolated');

GSSelect1=Greeny(:,1:(Ty/4));
GSSelect2=Greeny(:,(Ty/4):(Ty/2));
GSSelect3=Greeny(:,(Ty/2):(3*Ty/4));
GSSelect4=Greeny(:,(3*Ty/4):Ty);
GSSum1=sum(GSSelect1(:));
GSSum2=sum(GSSelect2(:));
GSSum3=sum(GSSelect3(:));
GSSum4=sum(GSSelect4(:));
if GSSum1>GSSum2 && GSSum1>GSSum3 && GSSum1>GSSum4
```

```
    fprintf('This apple is the leftmost apple\n')
else
    if GSSum2>GSSum3 && GSSum2>GSSum4
        fprintf('This apple is the second left apple\n')
    else
        if GSSum3>GSSum4
            fprintf('This apple is the third from left apple\n')
        else
            fprintf('This apple is the fourth apple from the left\n')
        end
    end
end
%
%
%
%
%
%
%
%
%
%
```

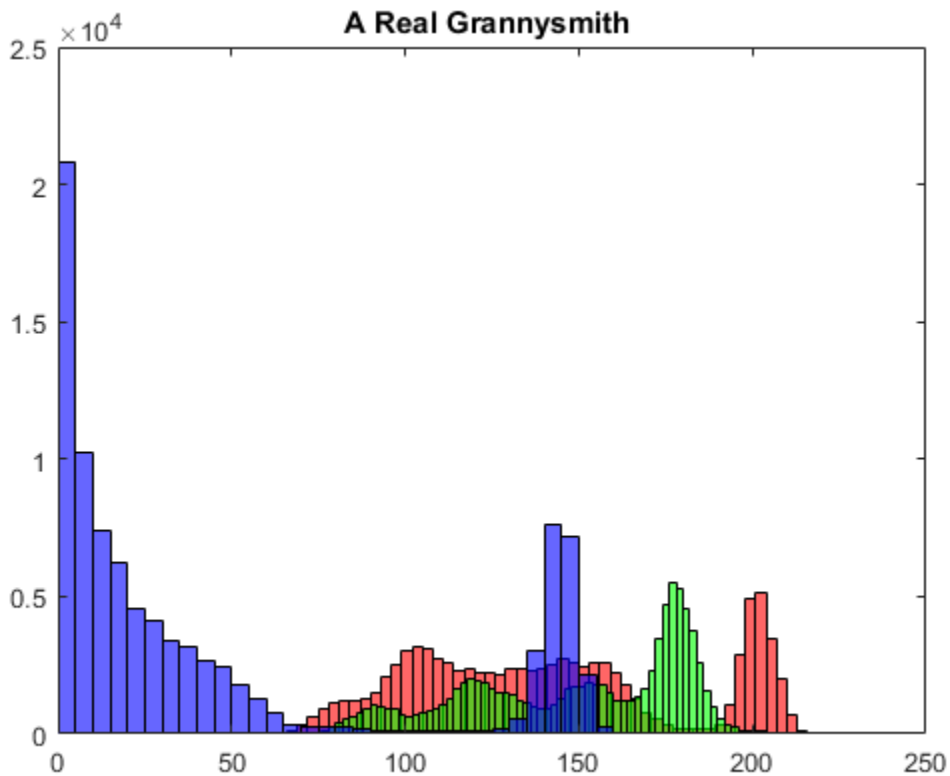
This apple is the leftmost apple



Part 5

```
Grannysmithreal= imread('grannyreal.jpg');
%imshow(Grannysmithreal);
Greenreal= Grannysmithreal(178:467,149:465,:);
R6=Greenreal(:,:,1);
G6=Greenreal(:,:,2);
B6=Greenreal(:,:,3);
figure; histogram(R6, 'facecolor', 'r')
hold; histogram(G6, 'facecolor', 'g'); histogram(B6, 'facecolor', 'b')
title('A Real Grannysmith ')
R7=Grannysmithreal(:,:,1); G7=Grannysmithreal(:,:,2);
 B7=Grannysmithreal(:,:,3);
Greenreal=uint8((G7>R7 & G7>B7 & R7>B7+25));
GSRShow=cat(3,Greenreal.*Grannysmithreal(:,:,1),Greenreal.*Grannysmithreal(:,:,2),
figure
subplot (1,2,2);
imshow(GSRShow);
title('Real Granny Smith Isolated');
subplot (1,2,1);
imshow(Grannysmithreal);
title('Real Granny Smith');
```

Current plot held



Published with MATLAB® R2017a